# Visual Hull Reconstruction via Space Carving

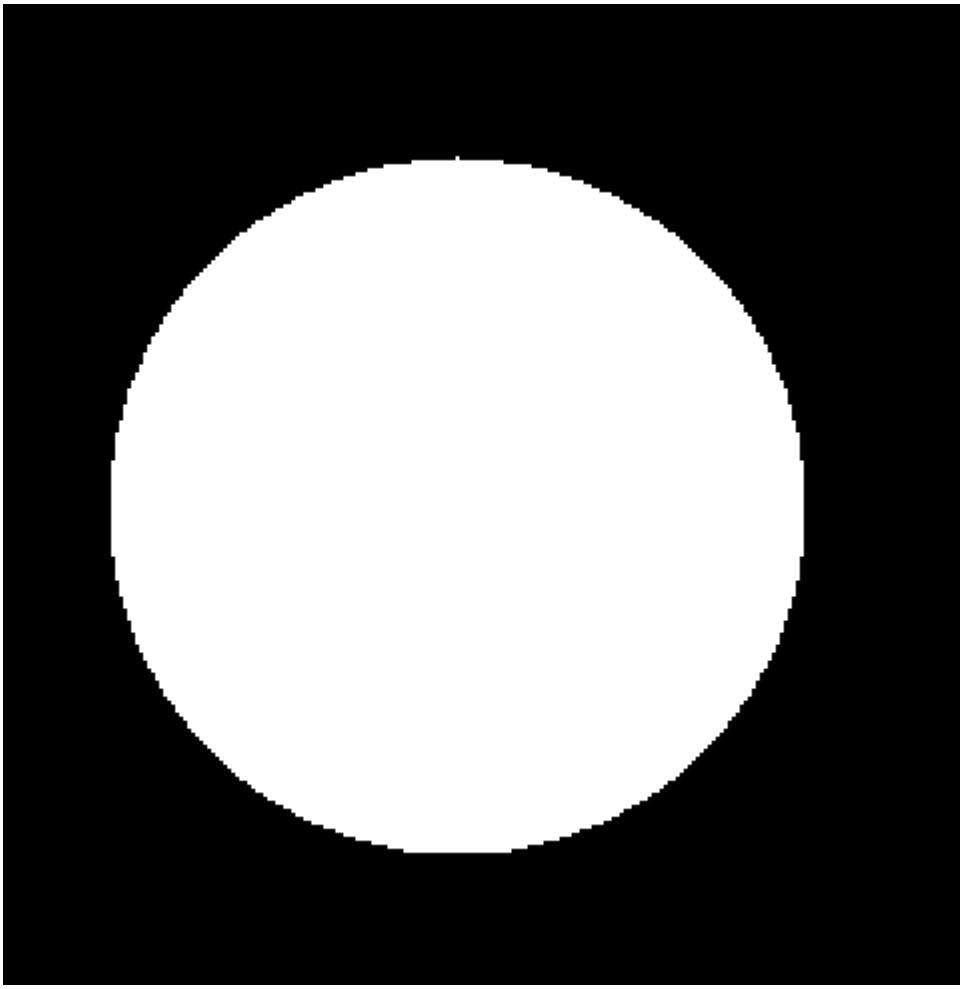
**Objective**
The objective of this assignment is to reconstruct the 3D shape of an object using multiple 2D views by applying space carving.

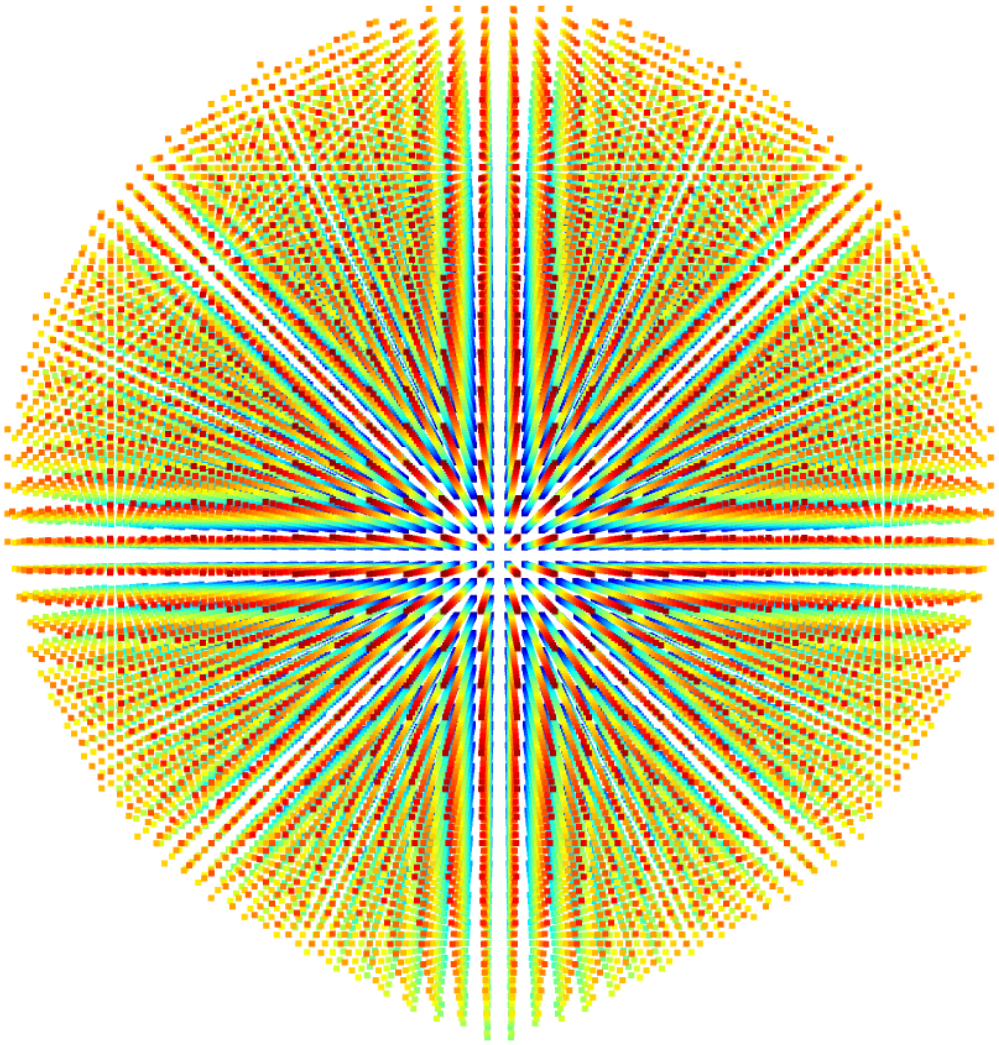
**Method**
The method follows a voxel-based space carving approach. First, a 3D mesh is either loaded or generated, and multiple virtual cameras are placed around the object. From each camera, a silhouette image is created by projecting the mesh onto a 2D plane. A voxel grid is then initialized to represent the possible 3D space of the object. Each voxel is projected into every camera view using the camera projection model. If a voxel projects outside the silhouette in any view, it is removed. This process continues across all views, gradually carving away empty space and leaving only the voxels that belong to the object's visual hull. The remaining voxels are then saved as a point cloud and can optionally be converted into a mesh using poisson surface reconstruction. This workflow is implemented in the provided code using libraries like Open3D and Trimesh

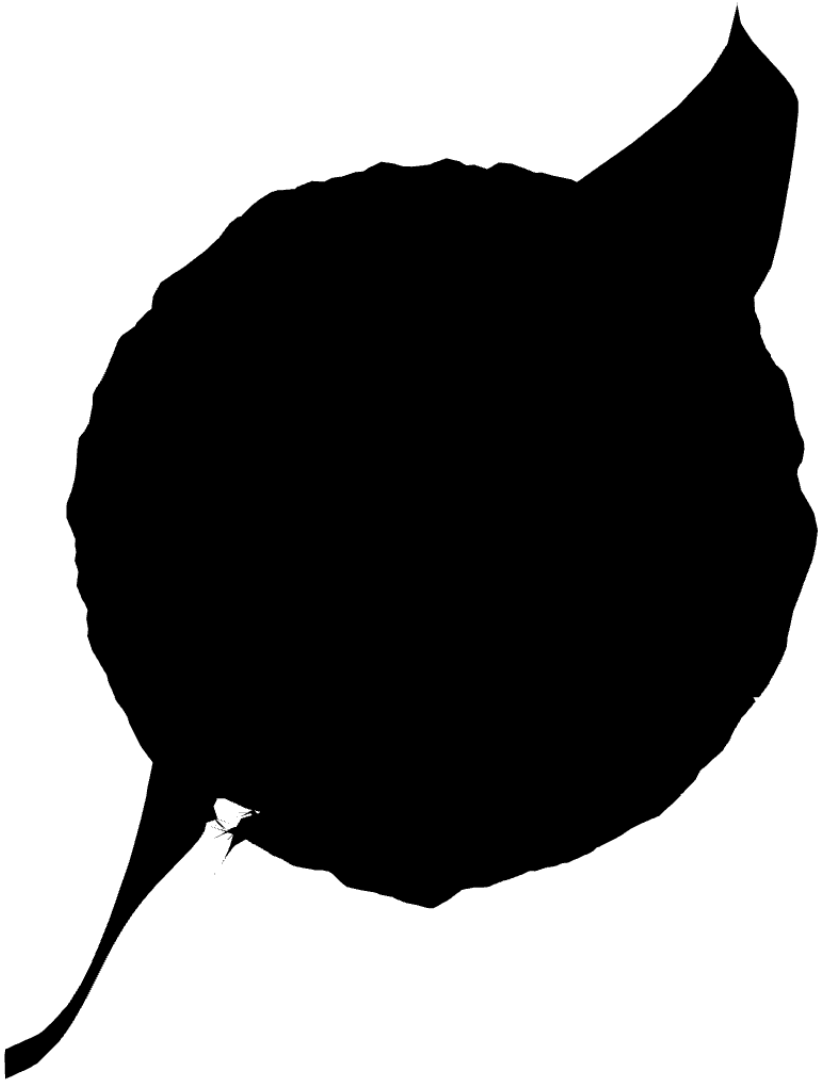
**Projection Model**
$x = K [R \mid T] X$
Where X is a 3D point, R and T are camera parameters (rotation matrix and translation vector respectively), K is the intrinsic matrix, and x is the projected pixel.

**Results**
Silhouettes, point clouds, and reconstructed meshes were generated for different view counts.

| Views | Resolution | Result |
|---|---|---|
| 5 | 48 | Rough visual hull with strong artifacts |
| 10 | 48 | Improved reconstruction |
| 20 | 48 | Clean and stable |
| 30 | 48 | Best result |







**Analysis**

Increasing the number of views improves reconstruction quality by reducing ambiguity in voxel carving. With only a few views, like 5, the visual hull is rough and contains many errors because there is not enough information to fully constrain the shape. As the number of views increases (10-30), the carving process becomes more accurate, and the final shape is much cleaner and closer to the original object.

**Limitations**

Cannot capture concave regions because silhouettes only provide outer boundaries. It is also sensitive to silhouette accuracy. Any noise or errors in silhouettes can lead to incorrect carving. Higher voxel resolution also improves detail but increases computation time and memory usage.

**Conclusion**

Space carving successfully reconstructs a visual hull and performance improves with more viewpoints.